ENGINEERING DESIGN & ARCHITECTURE PORTFOLIO

Multiplayer AI Arena: Gravitational Core

A high-performance tactile tactical game featuring A* Min-Heap pathfinding, DFS exploration, and multi-agent coordination.

OVERVIEW

Project Scope & Objectives

Core System Architecture

Algorithmic Intelligence

A* pathfinding with binary Min-Heap open sets and Euclidean distance heuristics for real-time tracking.

Cross-Platform Play

Supports up to 4 concurrent players with PC keyboard bindings and touch-optimized mobile D-Pad inputs.

Progressive Campaign

Five dynamic difficulty sectors featuring escalating obstacle densities and multi-agent sentinel formations.

AI SUBSYSTEMS

Pathfinding & State Machine Matrix

STATE 01

A* Min-Heap Pursuit

Optimized Euclidean distance evaluation tracking players across complex grid corridors.

STATE 02

DFS Exploration

Stack-based recursive wander patterns governing sentinel behavior during patrol phases.

STATE 03

Telegraph Vectoring

Predictive line-of-sight tracking projecting hazard lines prior to burst-dash maneuvers.

IMMERSIVE SYSTEMS

Quantum Core Mechanics & EMP Shockwaves

Ore Collection & Orbit Rings

Players extract scattered quantum ores, forming revolving orbit rings around hover craft.

EMP Stabilization Pulse

Depositing cores at the central shrine triggers a shockwave that stuns incoming sentinels.



DIFFICULTY SCALING

5-Sector Progressive Campaign

SECTOR 01

Reconnaissance

8 Quotas, 1 Sentinel

SECTOR 02

Neural Surge

12 Quotas, Symmetrical Corridors

SECTOR 03

Gravitational Rift

15 Quotas, 2 Sentinels

SECTOR 04

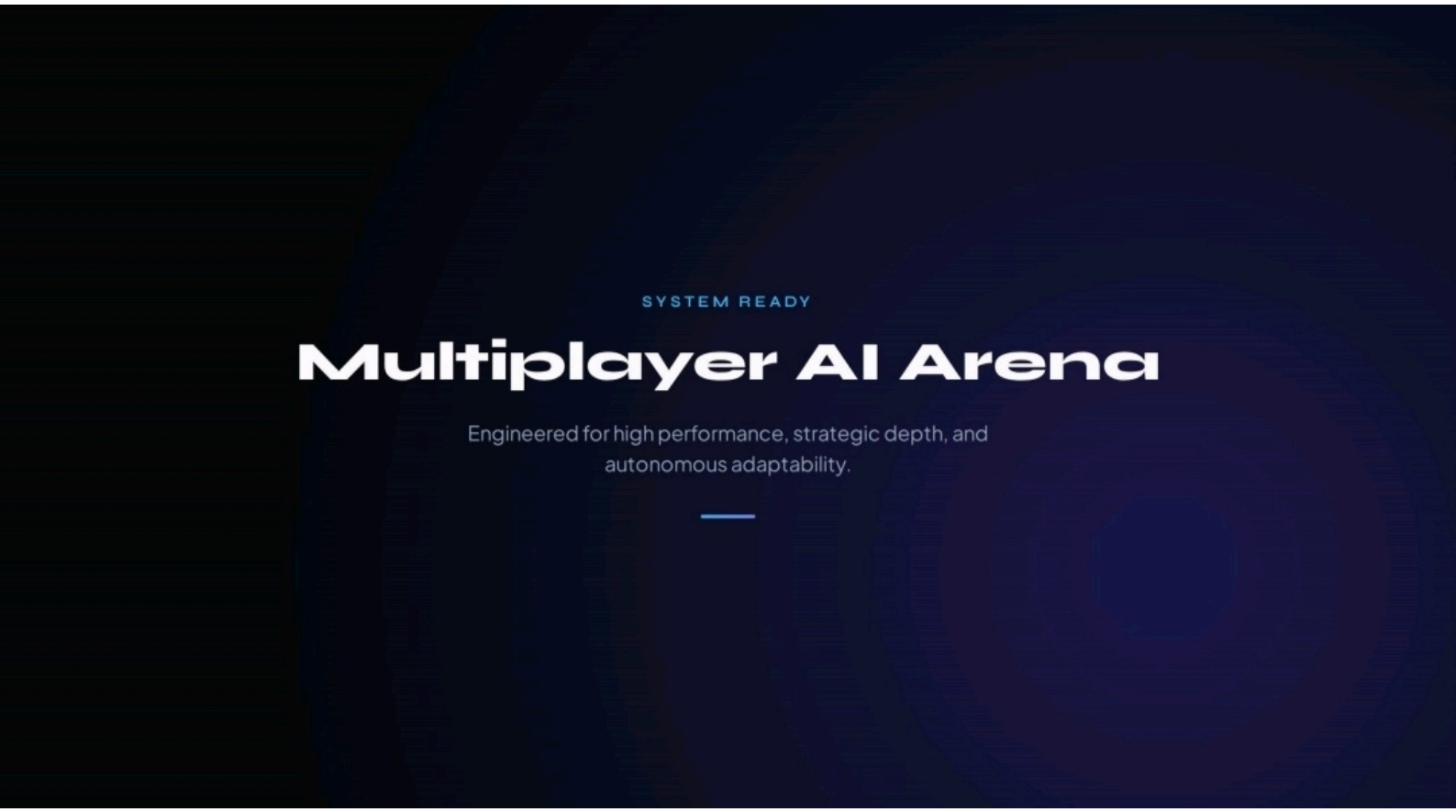
Overdrive Breach

18 Quotas, Labyrinth Maze

SECTOR 05

Apex Singularity

20 Quotas, 3 Sentinels


SYSTEM READY
Multiplayer AI Arena
Engineered for high performance, strategic depth, and autonomous adaptability.